

FLOOD TESTING GETS A REALITY CHECK

Although the structural damage to a home from flooding cannot always be prepared for, the use of flood-resistant materials can minimize the cost of rebuilding.

BY HESHMAT AGLAN AND ROBERT WENDT

Floods and flooding are catastrophic events brought on by spring thaws, thunderstorms, hurricanes, and tsunamis. One of the single largest contributors to the total cost of these events is the loss and replacement of residential housing. The unpredictability of the punching energy of these storms either in the form of water surge or wind force, along with the duration and depth of the flood and pollutants in the water, are just a few of the external factors affecting a building's ability to resist flood damage. The building's own design and materials also will strongly impact the resistance of a residential envelope to flood damage. Although the structural damage to a home from flooding due to initial impact stresses cannot always be prepared for, minimizing the damage associated with wetting through the use of flood-resistant materials is one method of reducing the cost of such events.

During the past five years, Tuskegee University and Oak Ridge National Laboratory (ORNL) have been conducting field and laboratory testing to study the resistance of building envelopes to physical degradation under wetting and drying cycles associated with flooding. We set out to identify materials and methods that can make the envelope of a house flood-damage resistant when used to repair houses after a flood or when building new homes in flood-prone areas. In the process, we developed a methodology to collect representative, measured, repro-



This homeowner and inspection team member are about to enter a flooded home. The flood level (scum lines) can be seen on the door and siding. Note the multiple distinct scum lines. These suggest that in this area draining occurred incrementally.

ducible—also known as scientific—data on how various residential materials and systems respond to flood conditions so that future recommendations for repairing flood damaged houses can be based on scientific data. The flooding that followed hurricane Katrina gave us ample opportunity to compare our findings from the simulated flood tests we performed at Tuskegee University with field observations from New Orleans. We are using the lessons we learned from this extensive natural disaster to further refine our recommendations.

Tuskegee Flood Test Facility

Obtaining building envelope characteristics during simulated flooding that closely correlate to actual flooding characteristics requires testing conditions that closely match what happens in real flooding. The Tuskegee Test Facility (TTF) was constructed to simulate actual flooding in entire building envelope systems (see “After the Flood—There’s Hope,” *HE* Sept/Oct ’04, p.18). Testing instrumentation was used to measure temperature and relative humidity of the environment as well as

the moisture content of the wall studs, wall surfaces, floor joists, and floor surfaces during the entire flooding and drying periods. Mold was sampled from the modules and tested in a laboratory to identify its type. Flexural strength and modulus of elasticity of various types of siding and wallboard were measured. Detailed protocols were used to guide visual observation, making them as detailed and consistent as possible throughout the series of tests. Extensive photographic records were also made.

Comparison of Flood Conditions

The flood conditions that homes were exposed to in New Orleans differed significantly from those the test modules experienced at the Tuskegee facility. For a start, the test module was flooded to a level of 2 feet above floor level while houses in New Orleans were exposed to flooding from several inches to more than 8 feet above floor level. How these different conditions will impact both the exterior and interior of a home isn't immediately apparent. For example, a gypsum board wall that was exposed to only 2 feet of uncontaminated floodwater for a short period of time is likely salvageable; however, it is not known if the entire wall were submerged over an extended period in contaminated floodwater if it will still be salvageable.

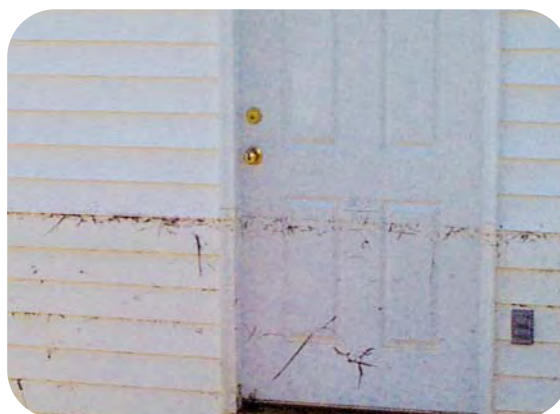
Here's what we found when we compared the two sets of results: The duration of flood exposure, water depth, and flooding and draining time can all impact the performance of houses. The delay before re-entry of building and natural drying conditions after flood exposure are also critical elements in determining both physical damage resistance and resistance to harboring microbes, bacteria, mold, and other contaminants. Understanding all of these variables is a formidable task.

Comparison of House Exteriors

In photographs of the outside of a New Orleans area home and a test module after flooding has receded, the water level, or scum line, in both instances can be clearly seen on the exterior of the walls



This is what a building exterior from a New Orleans home looked like after Katrina.



This is the building exterior of the Tuskegee Test module after flooding.

and doors (see photos). Similar debris remnants and staining can be seen on both units. The New Orleans home displays more staining, which is probably due to the longer flood duration and higher concentration of debris in the floodwater. Note that steel doors in both instances show minimal damage and appear to be able to be restored by washing.

Tuskegee tests showed that vinyl and fiber cement sidings both withstood flood conditions. Both vinyl and fiber cement siding could be restored to pre-flood conditions through washing the portion below flood level. They did not appear to have any adverse impact on adjacent materials in the wall system. Plywood sheathing maintained its integrity and mechanical properties; however it had not dried to pre-flood levels after 30 days. This was based on only three days of flood exposure. We do not know if this will be the case when the flood exposure time is extended to three or four weeks, especially when the plywood is in contact with fibrous batt insulation. Houses with vinyl siding in New Orleans showed little damage to the siding other than being dirty. Several gutted houses with vinyl siding were inspected and the sheathing, while still moist, appeared to have suffered only water staining.

In testing we found that windows and outside doors in most cases can be salvaged and reused. Double pane window glazing with baked enamel on aluminum frames withstood flooding exposure without deterioration. We also found that the foam filled steel exterior doors withstood multiple exposures to flooding and were able to be restored to pre-flood conditions with minimal effort. Wooden exterior doors performed satisfactorily during our test period, but these were new and had several coats of varnish. In New Orleans, double and single pane windows with vinyl and aluminum window frames appeared to survive the flood with no damage other than being dirty.

Solid wood panel exterior doors were swollen and frequently had to be kicked in to permit reentry. Some solid wood doors with a natural finish were water stained and many had their joints coming apart. Steel and fiberglass doors showed little deterioration other than being dirty. When these doors were installed in a wood frame they were ini-

tially hard to open due to the swelling of the frame.

We haven't tried to open wood windows in New Orleans, but it would make sense based on the swelling behavior of wooden doors and frames that they will swell and will probably be difficult to open.

Comparison of House Interiors

Monitoring of the flood test modules showed that moisture levels in wood studs that were above flood level returned to pre-flood levels within about 28 days. That portion of the studs and floor joists below the flood level dried to near the preflood moisture content, but had not achieved the preflood level during the test's 28-day drying period. We saw no visible evidence of mold growth on the wood framing when the test modules were autopsied. Wood framing is flood-damage resistant, as long as the wall or floor system of which it is a part can dry to normal levels within a reasonable timeframe of about one month or so. We believe that fiberglass or other moisture-retaining insulation in the exterior wall cavity or below subfloor may elevate the moisture level and significantly extend the drying period for the wood framing to the point that it becomes a durability concern.

In our 40-day tests of the moisture content and drying time of exterior wall and interior wall studs in a typical flood test module, we found some interesting patterns (see Figure 1). The rate of moisture absorption in the studs below water level, during flooding, is slightly higher for the stud in the exterior wall. The moisture content remained slightly higher during the entire flooding period. The moisture

content of the plywood subfloor, covered on the underside with fibrous insulation, remained higher longer than the other readings. Wood framing items adjacent to fibrous insulation—the exterior walls and floor joists—remained somewhat wetter than framing in inte-

but this appeared to have existed even before the flooding.

Fibrous batt insulation (fiberglass or rockwool) appears to retain moisture in the exterior wall cavities and below the subfloor. The moisture on the fibers appears to keep adjacent walls and floor materials wetter longer, which could potentially contribute to long-term damage to the subflooring, floor and wall framing, and the gypsum board walls. Most agencies recommend removal and replacement of fibrous insulation that has been subjected to floodwater.

When we tested spray polyurethane foam (SPUF) insulation in the wall cavities of the TTF, the wall board and wood studs in exterior walls dried at nearly the same rate as in the interior walls with empty cavities. SPUF absorbs water very slowly and was undamaged by flooding. SPUF did not retain moisture and as such did not have a negative impact on the flood damage resistance of the materials around it. Since it did not let water in to begin with no air flow was needed to dry it. In addition, there was no visible evidence of mold growth on the SPUF insulation. However, the performance of SPUF in long-duration flood events such as New Orleans has not yet been confirmed.

We measured the moisture content of gypsum wallboard before flooding, after reentry, and throughout the remainder of the drying period in the TTF. This was done on an interior wall and on the interior of a south wall at different elevations that ranged from floor level to six feet above the floor. We flooded the wallboard on day 2 of the test; day 5 is the draining of the basin; day 10 is the reentry into the module and the beginning of observation and measurement; and day 34 is the end of the drying period. For the gypsum board on an interior wall, it can be seen

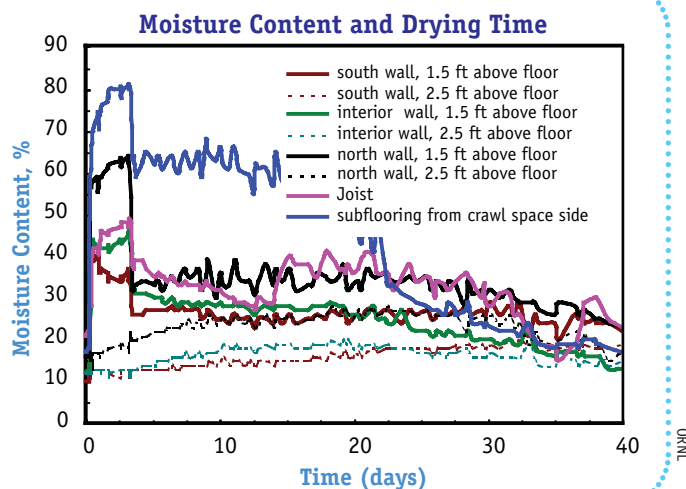


Figure 1. Day 1 through Day 4 is the period of flooding. The drying period begins with Day 4.

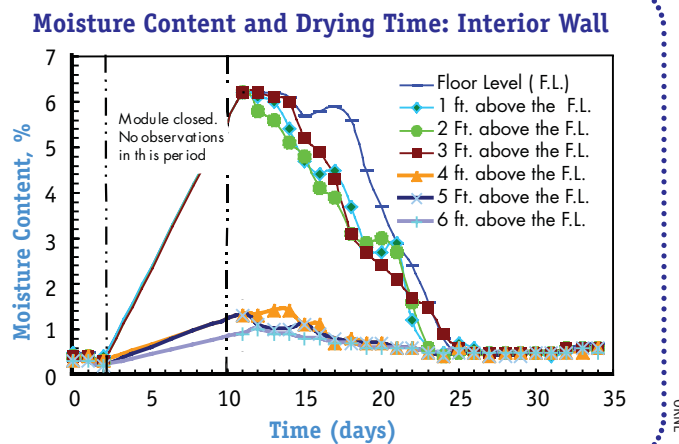


Figure 2. For the gypsum board on an interior wall, it can be seen that the maximum moisture content was about 6% at floor level, 1 foot, 2 feet, and 3 feet above floor level; 6% represents 100% saturation.

rior partitions that had no insulation. All framing returned to near preflood moisture levels within the test period.

A New Orleans house that had been gutted showed elevated moisture levels but otherwise no damage from flooding to the wall studs. There was no evidence of mold on most of the exposed wood framing. Minor mold growth was seen on a joist in one crawlspace,

that the maximum moisture content was about 6% at floor level, 1 foot, 2 feet, and 3 feet above floor level; 6% represents 100% saturation (see Figure 2). The moisture content at heights 4 feet and above was about 1% and reached the preflood conditions at the end of the drying period. It took about 15 days from the opening of the house for the moisture content at these levels to reach the preflood condition. Moisture content for the gypsum board cannot be compared with that of the wood products because of a difference in instrument calibration. All moisture contents given are relative to their original readings and not absolute values. The measurements taken at the floor level remained higher for about 10 days and then dropped rapidly.

In monitoring the moisture content for the gypsum board installed on an exterior wall facing South, we found that, similar to the interior wall, the highest moisture content was about 6% for the floor level, 1, 2 and 3 feet above the floor level (see Figure 3). The moisture content at heights 4 feet and above was just under 1% after opening the door and window and had reached the preflood condition at the end of the drying period. The floor level moisture also resembled that of the interior wall; however, it has a slower drying rate. This was the case for the floor level measurements and at 1 foot above the floor. For 2 and 3 feet above the floor the drying rate was faster.

The difference in drying rates between the interior and exterior walls at floor level is attributed to the impact of the fiberglass insulation in the exterior walls. The insulation retains more moisture at floor level. This increased moisture within the exterior wall cavity caused the exterior wall gypsum board to dry more slowly at the floor level. The drying rate at one foot above floor level is similar for both walls. At locations 2 and 3 feet above floor level, the exterior wall dried faster. This is attributed to the fact that this particular wall was facing South and absorbed heat from the sun.

When conventional paper faced gypsum board was used with fiberglass batt insulation on exterior walls, the gypsum board lost about 50% of its flexural strength and remained wetter than on interior walls (without insulation) at the end of the drying period. Interior gypsum board walls dried out and regained flexural strength during the test period. If gypsum board is able to dry completely within an appropriate time (about one month), it can be restored to preflood condition with only cosmetic restoration.

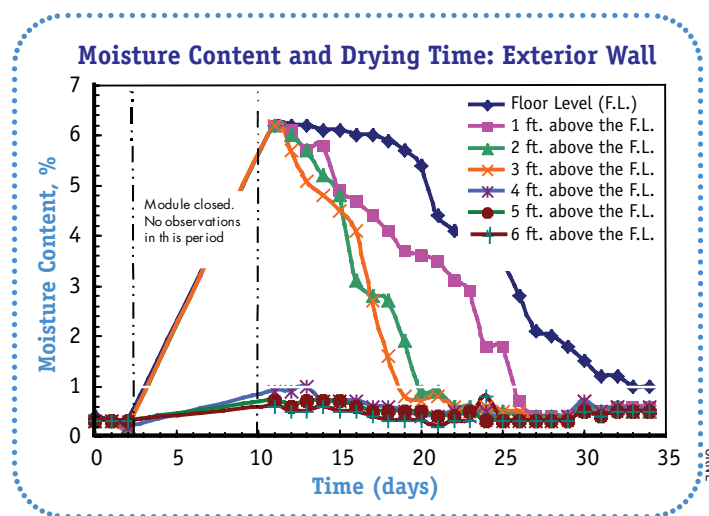


Figure 3. The difference in drying rates between the interior and exterior walls at floor level is attributed to the impact of the fiberglass insulation in the exterior walls. The insulation retains more moisture at floor level.

Observations from New Orleans showed that for the most part gypsum board and lath and plaster walls structurally survived the flooding. Many of these walls had significant mold growth and staining. Contaminants from the floodwater were also evident when samples of these materials were tested. The impact of fibrous insulation in exterior wall cavities was also very evident. Whether or not these walls in New Orleans could be restored has not yet been validated through testing. The current governing rule appears to be, "If it got wet, get rid of it."

Mold Growth

Wet surfaces encourage mold growth, which discolors surfaces, leads to odor problems, deteriorates building materials, and may cause allergic reactions and

other health problems in susceptible individuals. Moisture migration into building envelopes can occur via three main mechanisms. These are vapor diffusion, infiltration, and capillary action. Vapor diffusion and infiltration processes involve the transportation of water vapor. Infiltration of moisture into building envelopes is created by many factors, such as temperature gradient, wind, and air movement. Water vapor diffusion through porous building materials occurs due to the differential vapor pressure. Condensation can take place on building components where there is a change in permeability and where the surface temperature is below the dew point of the air. Moisture migration via capillary action occurs from flooding and/or a wet foundation. Saturated porous building material transports moisture vertically. (In this case, the gypsum board is submerged in water; therefore, there is no horizontal gradient.) This phenomenon is called "rising damp". Wet ground and

flooding provide the driving force for moisture migration by capillary action. High water tables, excessive rain, lateral water penetration through cracks, and leaking pipes can provide the necessary gradient for upward moisture migration, via capillary action or vapor diffusion.

In both the testing modules and in the New Orleans houses that we examined, we found that many building materials wick moisture, and it is on these surfaces that mold problems tend to start. Gypsum board, for example, if it is exposed to water can wick up to 20 inches in about 3 days. In the TTF, mold growth initially started on the surfaces of wicked moisture rather than on the submerged portion of the gypsum board after the water had receded. In a house in New Orleans, where the flood level was about five feet and the flood exposure was about three weeks, the

intensity of the mold was much higher. This is probably due to the longer exposure, the higher water level, the longer time to reenter the house to begin the drying process, and the nature of the debris and contaminants in the actual floodwater.

Fibrous insulation between the studs appears to have encouraged mold growth in one New Orleans home, probably by remaining moist for a longer period of time than surrounding materials, as evidenced by the vertical bands of mold growth. These bands appear to be reflecting the insulation in the wall cavities between the studs. We observed similar results in our testing in Tuskegee, although the intensity of the mold was not as high. Again, this is due to the differences in the flooding characteristics in each case.

Kitchen/bath base cabinets made of hard wood or fiberboard with a laminated finish can absorb moisture, experience swelling, and become severely stained due to flooding. This swelling can cause edge delamination on the doors and prevent closing of the cabinet doors. A cabinet in a test module at Tuskegee after flood testing at first showed no obvious material damage or mold growth. Initially, one might think about cleaning and refinishing these cabinets. However, closer examination of this cabinet revealed slight swelling and disfiguration. This caused the laminated surface to become debonded. While this material did eventually dry it was permanently disfigured and stained by the floodwater.

An equally significant impact was the fact that the base cabinet slowed the drying of the surrounding wallboard, and

these hidden areas supported mold growth. The cabinet trapped moisture between it and the adjacent walls, which led to the mold growth behind it. Some mold also grew on the uncoated portions



No mold or severe staining was observed in the TTF framing at autopsy, which occurred 15 months after the flood test.



Inspection of these exterior, insulated wall cavities in a New Orleans home revealed no damage to the studs and plywood sheathing. The fiberglass insulation was also undamaged. However, moisture levels of adjacent materials were elevated. The back side of the gypsum wallboard showed no evidence of mold growth.

of the base cabinet itself. We concluded that if the flooded cabinets are not removed shortly after the flood, that special provisions must be made to accelerate the drying of the adjacent walls and these areas should be visually inspected as part of the recovery process.

Field Testing Supports Lab Work

Our measurements at Tuskegee were often corroborated by our visual observations from New Orleans. Conventional paper-faced gypsum board wicked moisture during flooding and supported mold growth, both in our flood testing and in New Orleans. Many other conditions that we found in the lab were duplicated in the houses we inspected in New Orleans. Our observations indicate that scientific investigations of flood-resistant materials can guide the rebuilding or new construction process in flood-prone areas.

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FOR MORE INFORMATION:

For more on the study of flood resistant building materials, see *Field Testing of Energy-Efficient Flood-Damage-Resistant Envelopes* at http://www.ornl.gov/sci/res_buildings/NaturalDisaster.htm.